Exercise Sheet 15

The notation is that of Lecture 15. The letter ρ is the Weyl vector. The matrix of the action of H in a named basis is $D(H)$. Exercises 15.1–15.3 are routine calculations. Exercises 15.4–15.6 apply the lecture's proofs to new cases and rebuild their steps. Exercises 15.3, 15.4(iii), 15.6 and 15.7 use Verma modules or infinite formal sums. These formal sums use the lecture's completion $\mathcal{E}$ and, in Exercise 15.7, the opposite completion $\mathcal{E}^-$. Exercise 15.7 compares formal exponentials with functions on $\mathfrak{h}$.

Exercise 15.1 (Subtraction for $\mathfrak{sl}_2$). Write $z = e^{\omega}$, so $\operatorname{ch} V_m = z^m + z^{m-2} + \cdots + z^{-m}$.

(i) Compute $\operatorname{ch}(V_2 \otimes V_1)$ and decompose $V_2 \otimes V_1$ by subtracting characters.

(ii) Prove that $z^2 + z^{-2}$ is W-invariant with nonnegative coefficients, but is not the character of a finite-dimensional module.

Exercise 15.2 (The defining module of $\mathfrak{sl}_3$). Let $\mathfrak{h}$ be the diagonal traceless matrices and $\varepsilon_i(H) = H_{ii}$, so $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$ on $\mathfrak{h}$. Let $V = \mathbb{C}^3$.

(i) Express $\alpha_1, \alpha_2, \omega_1, \omega_2$ in the ε_i, compute $\operatorname{ch} V$ and check W-invariance.

(ii) Prove $\Lambda^2 V \cong V^*$ by characters.

(iii) Compute the coefficient of e^0 in $\operatorname{ch}(V \otimes V^*)$. Using $V \otimes V^* \cong \mathbb{C} \oplus \mathfrak{sl}_3$ from Lecture 14, find $\dim(\mathfrak{sl}_3)_0$.

Exercise 15.3 (Coefficient extraction in A_2). Write $p(a,b) = p(a\alpha_1 + b\alpha_2)$, with $p = 0$ if $a < 0$ or $b < 0$.

(i) Use the identity $\operatorname{ch} M(0) \cdot A_\rho = e^\rho$ to prove, for $(a,b) \neq (0,0)$,

$$p(a,b) = p(a-1,b) + p(a,b-1) - p(a-2,b-1) - p(a-1,b-2) + p(a-2,b-2).$$

(ii) Compute $p(3,2)$ by this recursion and by $\min(a,b) + 1$.

Exercise 15.4 (The B_2 denominator). Let the root system B_2 have simple roots α, short, and β, long, positive roots $R_+ = \{\alpha, \beta, \alpha+\beta, 2\alpha+\beta\}$, $\langle \beta, \alpha^\vee \rangle = -2$ and $\langle \alpha, \beta^\vee \rangle = -1$. Put $x = e^{-\alpha}$, $y = e^{-\beta}$.

(i) Expand $(1-x)(1-y)(1-xy)(1-x^2y)$.

(ii) Compute ρ and $W\rho$, and confirm the denominator identity.

(iii) Compute $p(2\alpha+\beta)$ and $p(2\alpha+2\beta)$. Write $p(a,b) = p(a\alpha+b\beta)$. Check the coefficients of x^2y and x^2y^2 in

$$\Big(\sum_{a,b\geq 0} p(a,b)x^a y^b\Big) \cdot (\text{the product in (i)}).$$

Exercise 15.5 (Anti-invariant elements). Call $f \in \mathbb{Z}[P]$ *anti-invariant* if $wf = \det(w) f$ for all $w \in W$.

(i) Prove $A_{u\nu} = \det(u) A_\nu$ for $u \in W$.

(ii) Prove $A_\nu = 0$ if $\langle \nu, \alpha^\vee \rangle = 0$ for some root α.

(iii) Prove that every anti-invariant f has a unique expression $f = \sum_{\nu \in \rho + P_+} c_\nu A_\nu$ with finitely many $c_\nu \neq 0$.

Exercise 15.6 (A difference realised by an exact sequence). For $\mathfrak{sl}_2$ and $m \in \mathbb{Z}_{\geq 0}$, let v_0 be the highest-weight vector of $M(m\omega)$ and v' the highest-weight vector of $M((-m-2)\omega)$.

(i) Write $\operatorname{ch} M(\lambda)$ for $\lambda = c\omega$, $c \in \mathbb{Z}$.

(ii) Let N be spanned by the $f^k v_0$ with $k \geq m+1$. Using Lecture 12, prove $N \cong M((-m-2)\omega)$, $M(m\omega)/N \cong V_m$, and

$$\operatorname{ch} V_m = \operatorname{ch} M(m\omega) - \operatorname{ch} M((-m-2)\omega).$$

(iii) Prove directly that

$$(e^{\omega} - e^{-\omega}) \operatorname{ch} V_m = e^{(m+1)\omega} - e^{-(m+1)\omega}.$$

Exercise 15.7 (Formal exponentials and analytic characters). For $H \in \mathfrak{h}$, define an additive map

$$\mathrm{ev}_H \colon \mathbb{Z}[P] \longrightarrow \mathbb{C}, \qquad \mathrm{ev}_H(e^\mu) = \exp(\mu(H)).$$

Here exp is the complex exponential, while e^μ is a formal symbol.

(i) Prove that ev_H is a ring homomorphism. Let $D(H)$ be the matrix of H in a weight basis of a finite-dimensional module V. Prove

$$\mathrm{ev}_H(\operatorname{ch} V) = \operatorname{tr} \exp D(H).$$

For $\mathfrak{sl}_2$, evaluate at $H = th$.

(ii) Prove that $f \in \mathbb{Z}[P]$ is determined by the function $H \mapsto \mathrm{ev}_H(f)$. Give an H for which the map ev_H is not injective.

(iii) Extend the evaluation to a formal sum $f = \sum_\mu f_\mu e^\mu$ with infinitely many terms by

$$\mathrm{ev}_H(f) = \sum_{\mu \in P} f_\mu \exp(\mu(H))$$

whenever this series converges absolutely. Leave $\mathrm{ev}_H(f)$ undefined otherwise. This evaluation is partial, even when restricted to $\mathcal{E}$. For $\mathfrak{sl}_2$, compare $\sum_{k \geq 0} e^{-k\alpha}$ and $g = -\sum_{k \geq 1} e^{k\alpha}$ formally and after evaluation at th.